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## **Enhanced Filtercake Removal Technology for Improved Operational Flexibility**

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### **Abstract**

This paper introduces a new generation of filtercake removal fluids that overcome the limitations of conventional systems used in the oil industry. Current practices involve using delayed-reaction cleanup fluids to homogeneously remove drilling mud filtercake from extended laterals. These systems are however capable of only operating within a narrow time frame of less than 24 hours which poses challenges, especially at elevated temperatures. This paper aims to address this constraint by presenting a novel dual-phase filtercake removal fluid system capable of delaying reaction for several days.

To achieve this goal, a comprehensive experimental program involving laboratory tests of the newly formulated emulsified filtercake removal fluid system was carried out at HPHT conditions. Specifically, the fluid was assessed for effectiveness in dissolving filtercake generated from water-based mud comprising of acid-soluble solids and polysaccharide polymers. The samples were subjected to varying soaking times ranging from 1 to 10 days. During this time, the samples were monitored for thermal stability via turbidity sensors, solubility, and measured their corrosivity on C-95 carbon steel metallurgy.

Extensive experimentation yielded encouraging outcomes. Notably, the dual-phase filtercake removal fluid demonstrated remarkable thermal stability at temperatures up to 210°F for 8 days. Moreover, it successfully dissolved increasing amounts of the filtercake from 24% to 51% during the respective soaking intervals, showcasing its ability to gradually break down complex deposits over an expanded timeframe. Additionally, lab investigations revealed minimal corrosivity toward typical tubular metallurgies, thereby alleviating concerns about equipment integrity. Collectively, these positive findings suggest that our innovation holds considerable promise for improving operational efficiencies and reducing downtime associated with current filtercake removal methodologies.

Unlike earlier approaches limited by short soaking times of no more than 24 hours, the proposed breakthrough formulation provides operators greater operational flexibility to conduct the necessary tasks in a safe manner. Thereby, potentially leading to increased resource recovery, reduced costs, and more streamlined workflows – all critical factors influencing project profitability and sustainability.

## Introduction

There is an increase in the drilling and utilization of multilateral wells to reap benefits like maximizing reservoir contact, enhancing sweep efficiency, and hastening oil recovery. These wells often also feature horizontal extended reach laterals, which pose special challenges for well intervention, especially in offshore environments where rig handling capacity is limited (Chishti, et al. 2024). Another substantial challenge is that reductions in well productivity due to formation damage is more pronounced in horizontal wells, due to factors like longer contact with drilling fluids (Al-Taq et al. 2014). This requires specially designed drilling fluids that minimize fluid loss to the formation through the creation of an external filtercake, and is commonly noted that only a small portion of the filtercake can be naturally removed through pressure drawdown (Burnett et al. 2002). Therefore, removal of filtercake through chemicals means is imperative to enhance well productivity as part of a comprehensive well completion strategy post-drilling operations.

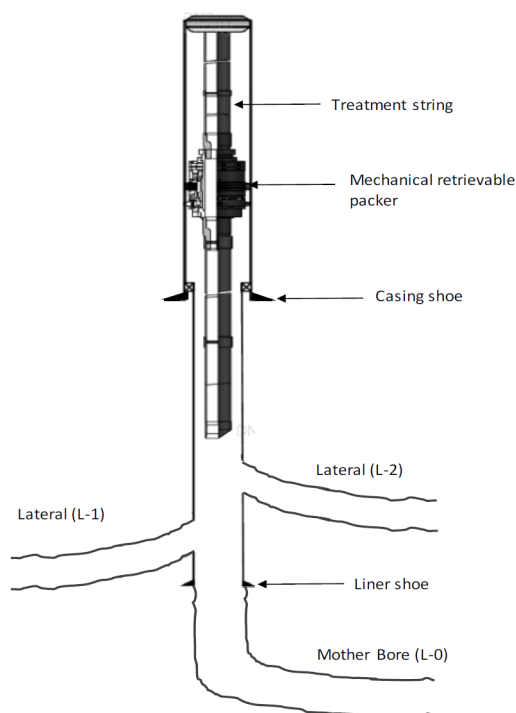
A typical water-based filtercake composition consists of xanthan, starch, and specially sized calcium carbonate particles. (Al-Taq et al. 2023). As such, typical chemical solutions have consisted of acids, oxidizers, and enzymes to dissolve the biopolymers and solids contained in the filtercake (Cobianco, et al. 2007). For an oil-based mud (OBM) system, an effective system must remove the oil from the cake surface, reverse the cake solid wettability, and dissolve the acid-soluble particles (Quintero, et al. 2007). The challenge with chemical removal is that many of these solutions are non-specific and results in filtercake dissolution in the vicinity of the injection area, leading to generation of loss zones, which ultimately take in the majority of the pumped acid. Utilizing fluids that delay the reaction such as acetate and formate esters (acid precursors) for in-situ acid generation, or the utilization of chelating agents, are commonly used methods for reaction control (Salgaonkar, et al. 2010). A successful chemical solution for filtercake removal must not only effectively remove the damage, but also delays the reaction in order to permit well completion operations without significant losses to the formation (Quintero, et al. 2007).

## Operational Sequence for Multilateral wells

Extended-reach multilateral wells offer significant advantages in reservoir access, thereby enhancing hydrocarbon recovery potential. However, executing effective filtercake removal treatments in these complex wells presents unique challenges, necessitating a specialized approach. A crucial factor in addressing these challenges is the design of a robust fluid system with precise reaction control, ensuring thorough filtercake removal while minimizing the risk of excessive losses.

The configuration of the well and the treatment objectives are paramount in determining the success of the filtercake removal process. Typically, multilateral wells feature multiple laterals (L-1, L-2, etc.) drilled from a central motherbore (L-0), utilizing either water-based or oil-based drill-in fluids. The primary objective is to remove filtercake damage caused by these fluids and restore permeability across each lateral. To achieve this, the treatment employs a reaction-controlled approach, which allows for an extended removal duration for a few days up to 10 days. This prolonged period ensures that necessary operations are completed without compromising well safety, thereby mitigating the risk of induced losses post-cleanup.

The deployment of coiled tubing (CT) through a temporary work string is a key component of this treatment strategy. This method provides a controlled and efficient means of removing filtercake and restoring permeability in each lateral, effectively addressing the complexities inherent in multilateral wells. Figure 1 below illustrates a multilateral wellbore with a temporary completion string, after displacing the drilling mud to completion brine.



**Figure 1—Example Well with a Temporary Treatment String for CT Deployment of Filtercake Removal System in Each Lateral.**

The CT is equipped with a multilateral tool (MLT) that enables access to each lateral, allowing for the precise placement of the treatment fluid. The treatment sequence typically involves running the CT with the MLT tool through the work string to access each lateral, spotting treatment fluids along the entire lateral section, and then pulling the CT out of the hole to surface. This process is repeated for each lateral, utilizing the reaction-controlled treatment that enables the necessary operations to take place without compromising well safety. The ability to pull out the CT to surface and redress the tool for subsequent runs allows for efficient and effective treatment of multiple laterals.

Following the filtercake removal treatment, it is essential to evaluate well performance to ensure production at optimal levels. This involves flowing back each lateral individually, for example, using nitrogen ( $N_2$ ) kick-off, before commingling the flow from all laterals together. Additionally, a production logging tool (PLT) should be run in the open hole section of the motherbore and across each lateral to gather data on the well's production profile. By following this operational procedure, operators can effectively remove filtercake damage and restore well productivity in multilateral wells, ultimately leading to improved well completion efficiency and increased hydrocarbon recovery.

## Extended-Delay Filtercake Removal System

Filtercake removal is based on the principle of dissolving or breaking down the filtercake, which is a complex mixture of solids and fluids. The filtercake is generated during the drilling process, as the drilling mud interacts with the crushed formation drill solids that forms a cake-like substance on the wellbore walls. In order to design an effective filtercake removal treatment fluid system, engineers must analyze the drill-in fluid (DIF) composition and formation lithology to identify the key filtercake components. A generic water-based formulation used to drill the multilateral wells described in this paper is shown in [Table 1](#) below.

**Table 1—Typical WBM Formulation used in Development of Extended Delay Filtercake Removal System**

| Generic Additive             | UOM | Typical Value |
|------------------------------|-----|---------------|
| Water                        | bbl | 0.851         |
| Biocide                      | gal | 0.05          |
| High Temp Fluid Loss         | lbs | 6 to 8        |
| Viscosifier                  | lbs | 1 to 2        |
| pH Adjustment                | lbs | 0.25 to 0.75  |
| Weighting Material           | lbs | 33 to 42      |
| Bridging Material – Size #1  | lbs | 5 to 10       |
| Bridging Material – Size #2  | lbs | 5 to 10       |
| Weighting Material – Size #1 | lbs | 5 to 10       |
| Weighting Material – Size #2 | lbs | 5 to 15       |
| Lubricant                    | %   | 1 to 3        |
| H <sub>2</sub> S Scavenger   | lbs | 0.5 to 3      |

The bridging and weighting materials are typically calcium carbonate-based, with the polymers being starch and xanthan gum. The dual-phase filtercake removal fluid system presented in [Table 2](#) below is based on a novel formulation that combines two phases: a solvent phase and a hydrocarbon phase. The solvent phase is designed to dissolve the filtercake, while the hydrocarbon phase helps to maintain the stability of the fluid over an extended period. The system is designed to operate at high temperatures and pressures, making it suitable for use in HPHT environments.

**Table 2—Generic Composition for the New Dual-Phase Extended Delay Filtercake Removal System**

| Generic Additive                                   | Typical Value       |
|----------------------------------------------------|---------------------|
| Completion Brine (e.g. NaCl or CaCl <sub>2</sub> ) | 50-60% by volume    |
| Organic acid blend/Acid Precursor/Chelant          | 3-8% by volume      |
| Retarding Agent                                    | 0.5-5% by volume    |
| Polymer Disruptor (optional)                       | 2-5% by volume      |
| Nanoparticles (optional)                           | 0.25-1.0% by weight |
| Emulsifier                                         | 3-10% by volume     |
| Diesel                                             | 20-35% by volume    |

The solvent phase is composed of a brine mixture, including enzymes, retarding agent, organic acids, esters and/or chelating agents. The use of esters to degrade and dissolve filtercakes is common with the use of water-based DIF systems ([Al-Otaibi et al. 2006](#); [Moajil et al. 2007](#); [Al-Mehailani et al. 2009](#)). Esters hydrolyze with time, temperature, and/or upon commingling with an aqueous fluid to yield an acid plus an alcohol or glycol. In particular, the use of glycol acid precursors can provide some delay as well as advantages that include the ability to vary the reaction rate and mitigate corrosion, emulsions and incompatibility with reservoir or completion fluids. Whilst the hydrocarbon phase is composed of diesel or a synthetic mineral oil with an emulsifier. Optionally, the oil external emulsion may be stabilized with silica-based nanoparticulates, which adsorb at the oil-water interface, enhancing stability and controlling the release of the solvent components. See [Figure 2](#) below for a photo of the Dual-Phase extended delay filtercake removal system.



Figure 2—Photo of Dual-Phase Extended Delay Filtercake Removal System

## Test Methodology

The experimental program conducted to assess the effectiveness of the dual-phase filtercake removal fluid system involved laboratory tests at HPHT conditions. The equipment used included high-pressure, high-temperature (HPHT) test cells, a HPHT filter press apparatus, a turbidity scanner, and a corrosion test apparatus.

To generate the filtercake used in the solubility experiments, a HPHT filter press apparatus was used with a generic WBM fluid. A certain volume of the WBM was placed in a HPHT cell fitted a ceramic disk saturated with a base brine. The cell is then pressurized, heated to the targeted temperature, and then the test was run with a differential pressure of 500 psi for 3 hours to generate the filter cake. For the solubility experiments, HPHT corrosion resistant test cells were used by placing a certain amount of filter cake and pouring the dual-phase removal system at a 1:10 cake to fluid ratio, heating to the targeted temperature, and running multiple tests for different soaking times, ranging from 1 to 10 days. The cell content is then filtered through a 0.45-micron filter paper and the remaining solids are weighted to determine the solubility percentage.

The Turbiscan is a device that measures the absorbance/transmissibility of the fluid, allowing for the determination of the thermal stability of the system due to coalescence rate of the emulsion as a function of time and temperature. This is a unique tool for monitoring the kinetics of destabilization of the system, and quantify it phenomena. The fluid is first prepared and heated to maximum unit temperature of 140°F, then subjected to the turbidity scanner for a duration up to 8 days. Lastly, the corrosivity of the system is measured by immersing C-95 carbon steel metallurgy coupons in the fluid, ensuring total coverage, and heating the fluid to a targeted temperature for the entire duration of the test. The coupon is weighed before and after the test to determine the corrosion rate via weight loss per unit surface area, and visually inspecting the coupon for signs of pitting.

## Filtercake Removal Efficiency Results

An initial filtercake removal assessment was performed using some traditional single-phase delayed systems, which included four variations of breaker systems and static soak periods ranging from 1 to 6 days at 200°F (see [Table 3](#) below). The results for these experiments are presented in [Tables 4](#) and [5](#) below.



Table 3—Traditional Single-Phase Filtercake Removal Systems

| Generic Additive                                   | Formulation #1 | Formulation #2 | Formulation #3 | Formulation #4 |
|----------------------------------------------------|----------------|----------------|----------------|----------------|
| Completion Brine (e.g. NaCl or CaCl <sub>2</sub> ) | x              | x              | x              | x              |
| Organic acid /Acid Precursor/Chelant               | x              |                | x              | x              |
| Retarding Agent                                    |                |                |                | x              |
| Polymer Disruptor (optional)                       | x              | x              |                |                |
| Surfactant                                         | x              | x              | x              |                |
| Nanoparticles (optional)                           |                |                |                |                |
| Emulsifier                                         |                |                |                |                |
| Diesel                                             |                |                |                |                |

Table 4—Initial Filtercake Dissolution Experiments @ 200°F

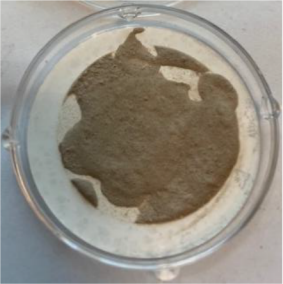
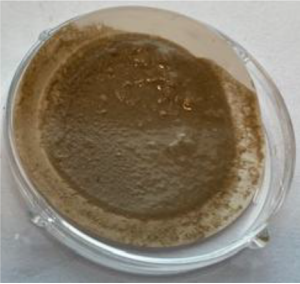
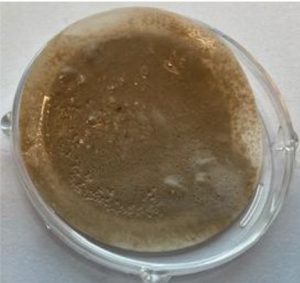
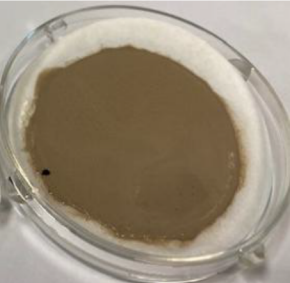
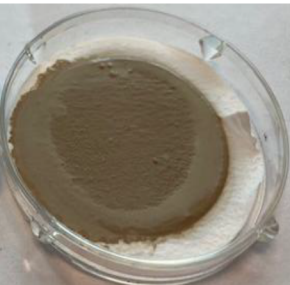
| Fluids         | Filtercake After 24 hrs                                                             | Filtercake After 3 days                                                              | Filtercake After 6 days                                                               |
|----------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Formulation #1 |    |    |    |
| Formulation #2 |   |   |   |
| Formulation #3 |  |  |  |
| Formulation #4 |                                                                                     |                                                                                      |  |

Table 5—Final Filtercake Dissolution for Traditional Single-Phase Filtercake Removal Systems

| Incubation Period (days) | Formulation #1 | Formulation #2 | Formulation #3 | Formulation #4 |
|--------------------------|----------------|----------------|----------------|----------------|
| 1                        | 35.9%          | 8.9%           | 26.3%          | NA             |
| 3                        | 60.4%          | 46.5%          | 53.3%          | NA             |
| 6                        | 69.4%          | 64.7%          | 75.2%          | 38.5%          |

Based on these initial results, some components of Formulations #2 and #4 were combined to form the solvent phase and used to prepare the Dual-Phase Extended Delay Filtercake Removal System. The solubility results are shown in Figure 3 below for a static soak period up to 8 days at a bottom temperature of 210°F with 2 different formulations (#5 and #6). Both of the fluids successfully dissolved increasing amounts of filtercake, from 24% to 51%, during the respective soaking intervals.

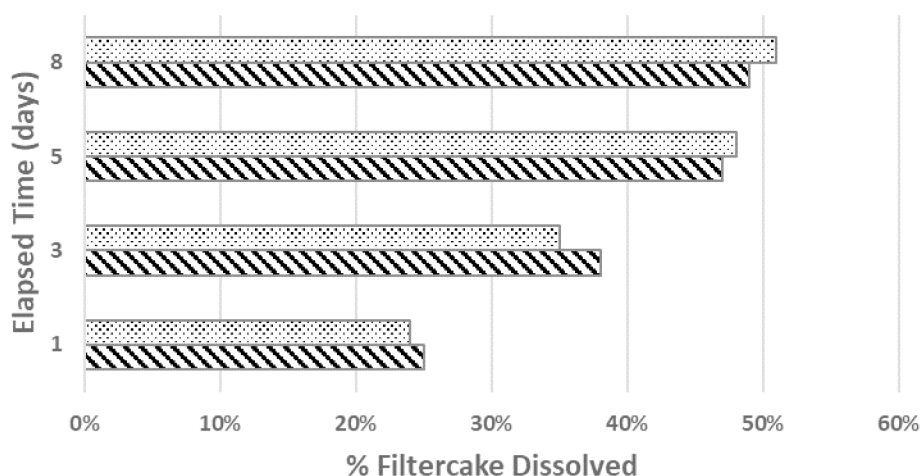
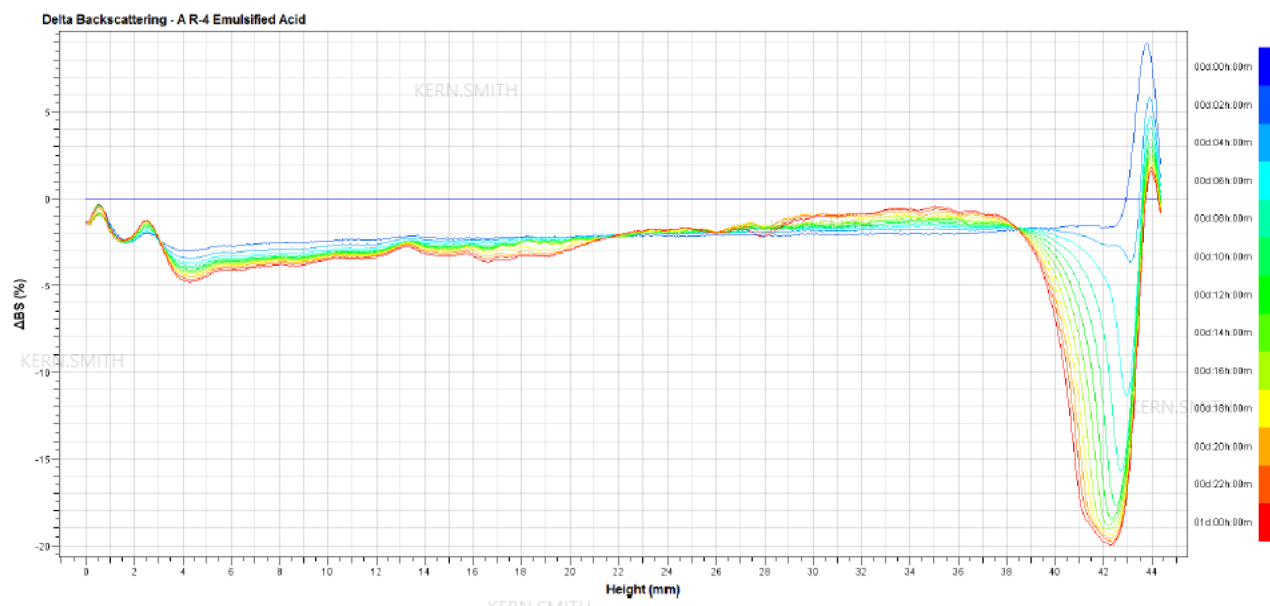


Figure 3—Filtercake Dissolution vs. Soaking Time

Below is an example result of the emulsion stability measurement over 1 day that shows the remarkable thermal stability of the dual-phase filtercake removal fluid, which extended over the next two days but with higher degree of destabilization over the remaining period.



The corrosivity of the fluid was minimal, with a corrosion rate of less than 0.05 lbs/square feet on C-95 carbon steel metallurgy.

**Table 6—Corrosion Rate of Formulation #5 on C-95 Coupon @ 210F**

| Incubation Period (days) | Corrosion Rate (lb/sqft) | Pitting Index |
|--------------------------|--------------------------|---------------|
| 1                        | 0.0001                   | 0             |
| 3                        | 0.0085                   | 0             |
| 5                        | 0.0200                   | 1             |
| 8                        | 0.0250                   | 1             |

## Conclusion

The development of the new dual-phase filtercake removal fluid represents a substantial advancement in addressing the challenges associated with filtercake removal in complex drilling operations, particularly in multilateral wells. The key findings of this study demonstrate that the novel fluid system exhibits exceptional performance under high-pressure and high-temperature (HPHT) conditions, maintaining stability up to 210°F for 8 days. This durability is complemented by the fluid's ability to dissolve filtercake, with a significant increase from 24% to 51% over a soaking period of 1 to 10 days. Additionally, the fluid's minimal corrosivity to steel ensures enhanced equipment longevity, reducing maintenance needs and potential downtime.

For practicing engineers, these findings offer several practical implications. The extended soaking times provided by this fluid system allow for greater operational flexibility, enabling more efficient planning and execution of drilling operations. Furthermore, the reduced corrosivity minimizes the risk of equipment damage, contributing to safer and more reliable drilling processes. These benefits collectively suggest that the new fluid system has the potential to enhance drilling efficiency, reduce operational costs, and improve resource recovery, ultimately leading to increased project profitability.

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## References

1. Al-Taq, Ali A., Habeeb H. Al-Haji, and Jaffar A. Saleem. 2014. First Successful Filtercake Damage Removal Treatment Utilizing In-situ. Paper presented at the SPE Annual Technical Conference and Exhibition, October 27–29, 2014. SPE-170832-MS. <https://doi.org/10.2118/170832-MS>
2. Al-Taq, Ali A., Hussain A. Al-Ibrahim, Ali A. Alsalem, and Abdullah A. Alrustum. 2023. Enzyme-Based Cleanup Fluid for High Temperature Filtercake Removal Applications. Paper presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, May 23–25, 2023. <https://doi.org/10.2118/214628-MS>
3. Burnett, David, Fernando Gutierrez, and Gerardo Serrano. 2002. Improved Filtercake Cleanup Practices for Horizontal Well Completions in Consolidated Sands: Improved Filtercake Cleanup Practices for Horizontal Well Completions. Paper presented at the SPE Annual Technical



- Conference and Exhibition, San Antonio, Texas, September 2002. SPE-77523-MS. <https://doi.org/10.2118/77523-MS>
4. Chishti, S. S., P. Correa, M. Alabdullah, and M. Ahsan. 2024. Maximizing Reservoir Potential: Coiled Tubing Multilateral Extended Reach Intervention for Acid Stimulation. Paper presented at the *ADIPEC*, November 4–7, 2024. SPE-222432-MS. <https://doi.org/10.2118/222432-MS>
  5. Cobianco, S., P. Albonico, E. Battistel, D. Bianchi, and M. Fornaroli. 2007. Thermophilic Enzymes for Filtercake Removal at High Temperature. Paper presented at the European Formation Damage Conference, May 30–June 1, 2007. SPE-107756-MS. <https://doi.org/10.2118/107756-MS>
  6. Quintero, Lirio, Thomas A. Jones, David E. Clark, and Allan Twynam. 2007. NAF Filtercake Removal Using Microemulsion Technology. Paper presented at the European Formation Damage Conference, May 30–June 1, 2007. SPE-107499-MS. <https://doi.org/10.2118/107499-MS>
  7. Salgaonkar, Lalit, Razien Ali, Achala Danait, and Sharifudin Salahudin. 2010. Strategies for Removal of Formate Brine-Based Drill in Fluid (DIF) Filtercake in Gravel-Packed Completions: A Laboratory Evaluation. Paper presented at the International Oil and Gas Conference and Exhibition in China, June 8–10, 2010. SPE-130199-MS. <https://doi.org/10.2118/130199-MS>
  8. Nasrallah, M., and Vinci, M. 2018. New Filter-Cake Breaker Technology Maximizes Production Rates by Removing Near-Wellbore Damage Zone with Delay Mechanism Designed for High Temperature Reservoirs: Offshore Abu Dhabi. Paper SPE-191888-MS presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition, Brisbane, Australia, 23rd October 2018. <https://doi.org/10.2118/191888-MS>
  9. Wagle, V., Al-Yami, A. S., Al-Abdullatif, Z., Bubshait, A.S. and Al-Safran, A. 2016. Operational Benefits of Designing Acid-Soluble Drilling Fluids for Horizontal Wells in HTHP Conditions. Paper presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, Abu Dhabi, UAE, January 2016. SPE-178214-MS. <https://doi.org/10.2118/178214-MS>
  10. Leschi, P., Demarthon, G., Davidson, E. and Clinch, D. 2006. Delayed-Release Acid System for Cleanup of Al Khalij Horizontal Openhole Drains. Paper presented at the SPE International Symposium and Exhibition on Formation Damage Control, Lafayette, Louisiana, USA, February 2006. SPE-98164-MS. <https://doi.org/10.2118/98164-MS>
  11. Al-Otaibi, M.B., Al-Moajil, A.M., and Nasr-El Din, H.A. 2006. In-Situ Acid System To Clean Up Drill-in Fluid Damage in High-Temperature Gas Wells. 2006. Paper presented at the IADC/SPE Asia Pacific Drilling Technology Conference and Exhibition, Bangkok, Thailand, November 2006. SPE-103846-MS. <https://doi.org/10.2118/103846-MS>
  12. Al-Otaibi, M.B., Siddiqui, M. A., and Nasr-El Din, H.A. 2004. Chemical Treatments to Enhance Productivity of Horizontal and Multilateral Wells: Lab Studies and Case Histories. Paper presented at the SPE/DOE Symposium on Improved Oil Recovery, Tulsa, Oklahoma, April 2004. SPE-89467-MS. <https://doi.org/10.2118/89467-MS>
  13. Al-Otaibi, M.B., and Nasr-El Din, H.A. 2005. Chemical Treatments for Removal of Drill-in-Fluid Damage in Horizontal-Multilateral Gas Wells: Lab Studies and Case Histories. Paper presented at the SPE Europe/EAGE Annual Conference, Madrid, Spain, June 2005. SPE-94043-MS. <https://doi.org/10.2118/94043-MS>
  14. Nasr-El Din, H.A., Al-Moajil, A.M., and Al-Aamri, A.D. 2007. Evaluation of In-Situ Generated Acids for Filter-Cake Cleanup. 2007. Paper presented at the European Formation Damage Conference, Scheveningen, The Netherlands, May 2007. SPE-107537-MS. <https://doi.org/10.2118/107537-MS>
  15. Harris, R.E., McKay, I.D., Mbala, J.M. and Schaaf, R.P. 2001. Stimulation of a Producing Horizontal Well Using Enzymes that Generate Acid in situ – Case History. Paper SPE-68911-MS

- presented at the European Formation Damage Conference, The Hague, The Netherlands, 21–22 May 2001. <https://doi.org/http://dx.doi.org/10.2118/68911-MS>.
16. Marion, D., Noman, R., Al-Othman, I. J. and Leschi, P. 2003. The development of Al Khalij - A complex carbonate field, offshore Qatar. Paper SPE-81574-MS presented at the SPE Middle East Oil Show and Conference, Bahrain 5–8 April 2003. <https://doi.org/http://dx.doi.org/10.2118/81574-MS>.
  17. Davidson, E., Mota, L., Mosley, N., Chimara, G., Morrison, A. and Archibald, I., 2012. New and Effective Filter cake Removal Optimizes Water Injectivity. Paper SPE-151683-MS, presented at the SPE International Symposium and Exhibition on Formation Damage Control, Lafayette, Louisiana, USA, 15 – 17 Feb 2012. <https://doi.org/http://dx.doi.org/10.2118/151683-MS>.
  18. Almond, S. W., Harris, R. E. and Penny, G. S. 1995. Utilization of Biologically Generated Acid for Drilling Fluid Damage Removal and Uniform Acid Placement across Long Interval. Paper SPE-30123-MS presented at the European Formation Damage Conference, The Hague, The Netherlands 15–16 May 1995. <https://doi.org/http://dx.doi.org/10.2118/30123-MS>.
  19. L. S. Kim, L.S., Ravitz, R., Patel, A., Martens, H., Luyster, M. and Kuck, M. 2011. Specially Formulated Delayed-Breaker System for Extended-Reach Injectors in a Viscous Oil Waterflood Application. Paper presented at the SPE European Formation Damage Conference, Noordwijk, The Netherlands, 7 June 2011. SPE-144136-MS. <https://doi.org/10.2118/144136-MS>